



PROFESSIONAL SUMMARY

Entry-level AI & Full-Stack Developer with an MCA and B.Sc. in Computer Science. Hands-on with full-stack and AI-powered applications, plus a few blockchain projects, using Java, Python, React, Node.js, Flask, and LLM APIs (Groq). Built and shipped 8+ end-to-end projects independently, backed by certifications in SQL, Data Analysis, Prompt Engineering, and Anthropic's Claude/MCP tools. Looking for Software Developer, Full-Stack Developer, or AI Engineer roles, available immediately and open to relocating anywhere in India.

EDUCATION

MCA | Avanthi Institute of Engineering & Technology (Affiliated with JN-TUH)
2024 – 2026 | Status: Completed
Hyderabad

B.Sc. Computer Science | Vivekavardhani Degree College, Kothagudem (Affiliated with Kakatiya University)
Completed 2024 | Warangal | CGPA: 6.51/10

Intermediate & 10th | Sri Nalanda Jr. College (2019) & Singareni Collieries High School (2017)
CGPA: 6.32/10 & 6.7/10 | Kothagudem

EXPERIENCE

Academic & Freelance Project Experience

Independent Software & AI Developer 2024 – Present

- 8+ applications built and deployed from scratch, spanning full-stack development, AI integration, and blockchain work, with database design, REST APIs, and security built into each one
- Used Claude and Groq in the dev workflow to prototype faster, debug quicker, and ship sooner

TECHNICAL SKILLS

Languages: Java (OOP, Java 8+), Python, JavaScript, SQL, PL/SQL, C++

AI & LLM: Groq API, LLM Integration (LLaMA), Prompt Engineering, Cloud API Integration

Web Development: React.js (Vite), Node.js, Express.js, Flask, JSP, Servlets, HTML5, CSS3, REST API Design

Database: MySQL, PostgreSQL, CRUD Operations, Database Design

Architecture & Tools: Software Architecture, Fault-Tolerant System Design, Apache Tomcat, Git, GitHub, Docker, Postman

Security & Concepts: Counting Bloom Filters (CBF), Cryptographic Verification, Data Security & Encryption, Public Verifiability

CERTIFICATIONS

- SQL and Relational Databases 101
- Data Analysis Using Python
- Prompt Engineering for Everyone
- Deloitte:** Cyber Security Simulation, Data Analytics Simulation
- Anthropic:** Claude 101, Claude Code 101, Claude Code in Action, Claude with the Anthropic API, Introduction to MCP, MCP Advanced Topics
- Udemy:** Complete Prompt Engineering for AI Bootcamp (2026), Complete Full-Stack Web Development Bootcamp

KEY STRENGTHS

- AI-Assisted Efficiency:** Claude, Groq, and prompt engineering keep prototyping and development moving fast
- 8+ projects, built independently and deployed one by one, alongside MCA coursework
- Most of the real problem-solving came from debugging deployment issues and getting APIs to actually talk to each other
- Quick learner. Self-taught new technologies (Groq API, Motoko/ICP, RAG) through independent research
- Cares about writing code that stays clean and scales, and that someone else could maintain without a headache

PROJECTS

AI Projects

Panda AI - Multi-User AI Assistant Platform

Flask, Python, Groq API, REST APIs

A text chatbot wasn't enough. I wanted something with real web search and file understanding, so I built this instead of relying on a single paid API.

- Multi-user AI assistant with real-time search grounding and multi-modal file processing: PDF, image, audio, video
- Fallback architecture across 3 Groq API keys plus OpenRouter, so it stays up and costs nothing to run
- Kept the REST API stateless, so multiple users can work at once without any drop in performance

Universal AI-Powered Resource & Budget ERP System

Java, Groq LLM, REST API

AI can do more than run a chatbot. This explores how it can support real business decisions, like knowing when to reorder stock or flagging a suspicious transaction.

- Multi-domain ERP system with an AI engine handling business insights, auto-reordering, and fraud detection
- Built the REST API and browser dashboard to support concurrent multi-user sessions across 4+ business modules

Java Projects

Secure Cloud Migration & Verifiable Deletion System

Java 8, Counting Bloom Filters, Cryptographic Verification, MySQL, Apache Tomcat

This was my MCA major project at Avanthi Institute, built to solve a real cloud security problem: migrating data safely between providers while guaranteeing permanent, verifiable deletion, without relying on a third party.

- Cloud-to-cloud data transfer and verifiable deletion framework, built on Counting Bloom Filters (CBF), which removes the need for a Trusted Third Party (TTP)
- Added cryptographic confidentiality, integrity validation, and public verifiability protocols so malicious server activity can actually be caught

Inventory Management System

Java, Java Swing, ArrayList, OOP

Built this to understand how businesses actually track profit and stock in real time, and to get hands-on with Java Swing for the first time.

- GUI built in Java Swing, with real-time data shown through JTables
- Profit and stock calculations run automatically, backed by a Low Stock Alert and transaction logging

Web Development Projects

WaveCoin - Decentralized Token Application

Motoko, Internet Computer (ICP), DFX

Blockchain always felt like a black box, so I built this to see how minting and transfers actually work under the hood, instead of just reading about crypto.

- Custom cryptocurrency logic for minting and transferring tokens on-chain
- Deployed the canisters and built a frontend for real-time balance tracking

Advanced React Blog Platform

React.js, Node.js, Express, PostgreSQL/MongoDB

Basic CRUD wasn't enough of a challenge, so this became a deeper dive into real authentication and session handling in a full-stack app.

- Full CRUD app, with secure authentication and session handling built in

Additional Projects

Also built an AI Blog Platform with AI content generation and a RAG chatbot, using React and the Groq LLaMA API. There's also an E-Commerce Console App in Java and MySQL that simulates a full shopping and wallet system. And PeakNFT, an NFT minting platform built with React.js and Solidity/Motoko, using decentralized storage for metadata. Full code for all of these is on GitHub at github.com/Upendrasai1101.